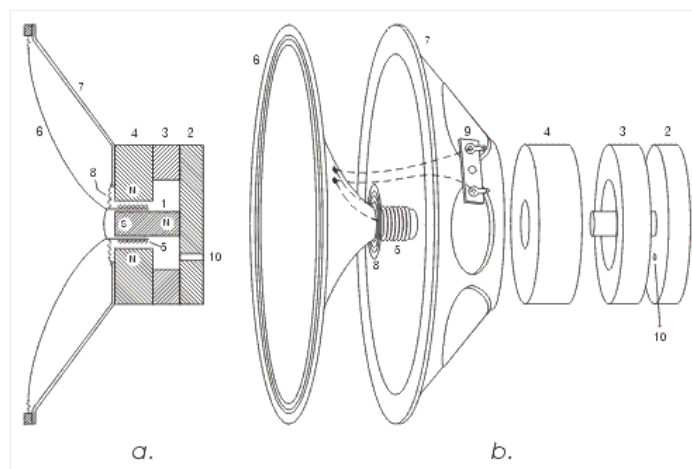


8.2 SPEAKERS

MIKROELEKTRONIKA

Speakers vary enormously in size and shape. They can be designed as crystal or capacitive, but most often they are dynamic (called electro-dynamic construction).



The cross-section of an electro-dynamic speaker is shown in 8.4. Ferrite rings (2, 3 and 4) are added to a large permanent magnet (1) which creates a strong magnetic field in the narrow gap between magnets North and South poles. A Cylindrical former is added to the gap and it holds coil (5). The ends of the coil are taken to the outside of the speaker.

The two most important characteristics of a speaker are its resistance (we actually call the resistance of a speaker **IMPEDANCE** as the value is determined at a frequency of 1kHz and the value is higher than its actual resistance) and its wattage. Common impedances are 4, 8 and 16 ohm, but there are also 1.5, 40 and 80 ohm speakers. Speaker wattages range from a fraction of a watt to hundreds of watts.

When choosing a speaker, it is advisable to choose the largest speaker possible as they are more efficient and produce the least distortion especially in the low frequency range. Speakers should be housed in a large box since it functions as a resonating chamber and this greatly adds to the overall quality of the sound.



